Agent-Based Monte Carlo Simulation of Tumor Growth

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1 Introduction

In silico simulations are increasingly used to study tumor evolution, growth, and progression, as they allow one to explore complex biological processes in a controlled and systematic way. Such approaches are particularly useful when spatial effects, local interactions, and stochasticity play an important role, as is the case in solid tumor growth. In this project, tumor growth is modeled using an agent-based Monte Carlo approach. Agent-based models (ABMs) describe a system in terms of individual cells that follow simple, local rules depending on their state and immediate surroundings. In cancer growth modeling, ABMs are well suited to capture spatial heterogeneity and emergent behavior arising from cell–cell interactions and environmental constraints. A common formulation employs on-lattice models, where cells occupy sites on a regular grid and interact with neighboring cells.

The aim of this work is to create a model that can be used to study the impact of stochasticity and spatial constraints on tumor growth taking into account the oxygen avalibility, probability of cellular division and death. The stochastic nature of cellular processes is simulated using MOnte Carlo methods implemented in Julia programming language, in which events such as division, death, and mutation-site occur probabilistically at each time step.

2 Model and Methods

The tumor is simulated on a two-dimensional square lattice of size $N \times N$, where in this work $N = 301$. Each lattice site can be empty or occupied by a normal cell, a cancer cell, or a dead cell. The system is initialized with a single normal cell placed at the center of the lattice. Time evolution proceeds in discrete steps using a Monte Carlo update scheme. At each time step, all living cells are visited once in random order, and stochastic decisions for division or death are made taking into account the local oxygen availability.

A mutation event is introduced by converting a normal cell that has at least one neighboring empty lattice site into a cancer cell.

The mutation time T_{mut} is controlled to explore different growth scenarios, and simulations were performed using different random seeds to generate independent realizations of the stochastic process.

Division and death probabilities depend on both cell type and local oxygen concentration, with the effective parameter values summarized in Table 1. Cancer cells are assigned a higher division probability and a lower death probability compared to normal cells, reflecting their increased proliferative capacity and assumed tolerance to hypoxic conditions.

Table 1: Effective probabilities used in the simulations.

Parameter	Normal cells	Cancer cells
Division probability p_{div}	0.30	0.40
Death probability p_{death}	0.05	0.02

The oxygen field is prescribed as a static, radially symmetric profile that is lowest at the lattice core and increases toward the lattice boundary, mimicking diffusion-limited oxygen supply. It is defined in Equation (1).

$$O(r) = O_{\min} + (1 - O_{\min}) \left(1 - e^{-r/L}\right), \tag{1}$$

where r is the distance from the lattice center, $O_{\min} = 0.35$ denotes the minimum oxygen level, and L controls the characteristic length scale of oxygen recovery.

Cell division is attempted by placing a daughter cell in a neighboring empty site. If no empty site is available, division may still occur through a pushing mechanism, in which a chain of neighboring cells is displaced along a lattice direction until an empty site is reached. Cancer cells are assigned a higher probability to push neighboring cells compared to normal cells, reflecting their enhanced mechanical aggressiveness. For normal cells, the pushing probability is set to a very small value ($p_{\mathrm{push}} = 0.05$) and is therefore negligible in practice. Dead cells are removed stochastically with a fixed clearance probability $p_{\mathrm{clear}} = 0.02$, allowing space to be freed over time.

3 Results and Discussion

3.1 Effect of mutation timing on tumor growth

In figure 1, the effect of mutation timing on tumor growth is illustrated for seed 6. The left panel shows the lattice snapshot at the mutation time t = 50, 90, 150 and 200. $O_{2\mathrm{MUT}}$ is the local oxygen concentration at the mutation site. The mutation site is marked by a yellow circle.

These snapshots illustrate how the timing of the mutation influences the spatial competition between cancer and normal cells. When the mutation occurs at an earlier time, the cancer population has more time to expand, resulting in stronger crowding and a clear dominance over the surrounding normal cells. In contrast, for later mutation times, normal cells have already proliferated and occupied a larger fraction of the lattice before the mutation takes place, leading to a more abundance of normal cells.

Interestingly, while earlier the mutation happened at the periphery, at $t = 200$, the mutation occurs at more centrally located site. This situation arises from the stochastic clearance of dead cells implemented in the model, which free up an empty site for cells to divide into. Cancer growth in this case remains significantly reduced compared to earlier mutation times, highlighting the importance of early mutation events and spatial constraint in driving tumor expansion.

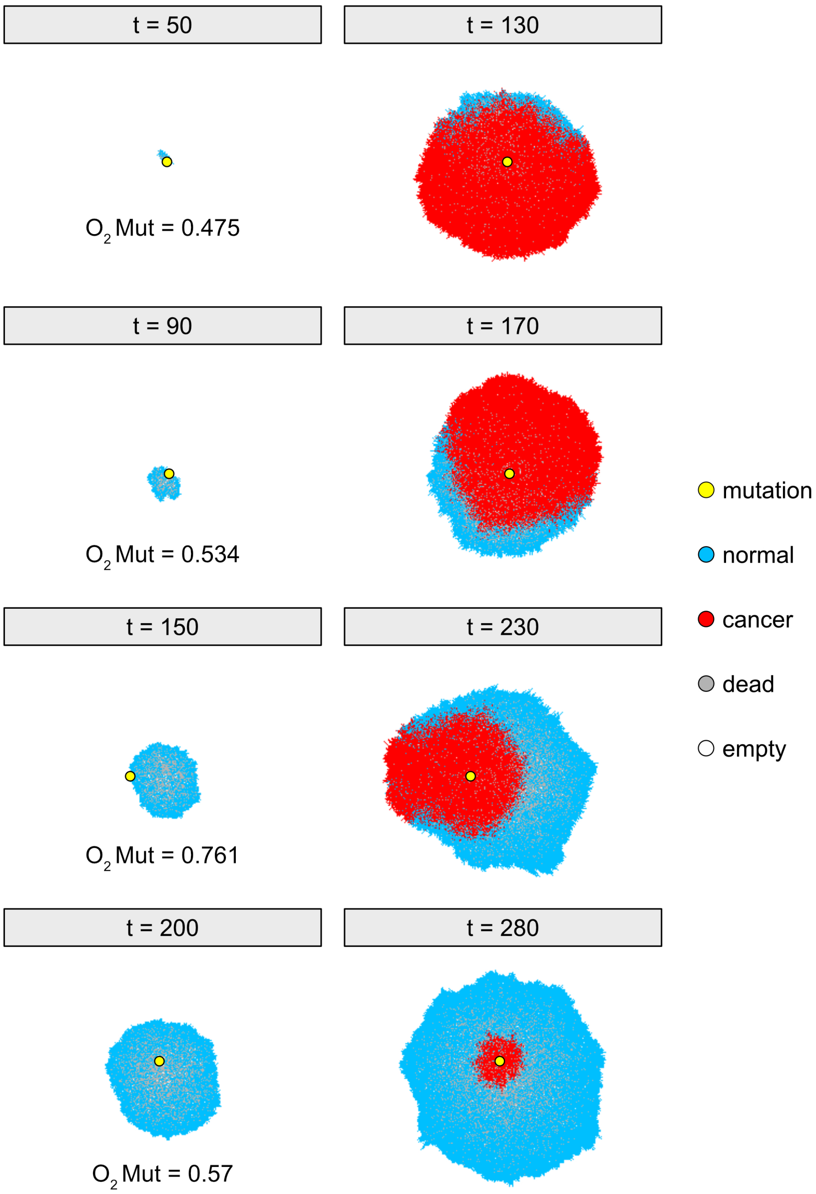

Figure 1: Agent-based lattice snapshots illustrating the impact of the mutation event on tumor growth. Left panels show the lattice configuration at the time the mutation is introduced, while right panels show the corresponding configurations after a fixed interval of 80 time steps following the mutation. t denotes simulation time, the yellow circle marks the mutation site, and O_2^{mut} denotes the local oxygen concentration at that site.

3.2 Effect of crowding on cells growth

To investigate the role of stochasticity in tumor evolution, simulations were performed with a fixed mutation time for different seeds. Figure 2 shows representative lattice snapshots for seeds 6 and 10 at the mutation time $t_{\mathrm{mut}} = 120$ (left column) and at a later time $t = 200$ (right column).

Even with the same mutation time, the outcomes differ strongly between seeds. In seed 10, the cancer population expands rapidly and occupies most of the lattice by $t = 200$, whereas in seed 6 the cancer growth remains limited. This difference originates from the stochastic placement of the mutation, which determines the local environment in which the first cancer cell appears.

Further analysis shows that oxygen availability at the mutation site is not the main factor behind this behavior. Instead, the key difference is the local density of normal cells at t_{mut}. Quantifying this through the number of normal cells, we find $N_n(t_{\mathrm{mut}} = 120, \mathrm{seed6}) = 1225$, compared to $N_n(t_{\mathrm{mut}} = 120, \mathrm{seed10}) = 288$. The much lower crowding in seed 10 allows the cancer population to expand more easily, especially given its higher division and pushing probabilities and lower death probability compared to normal cells, as described earlier.

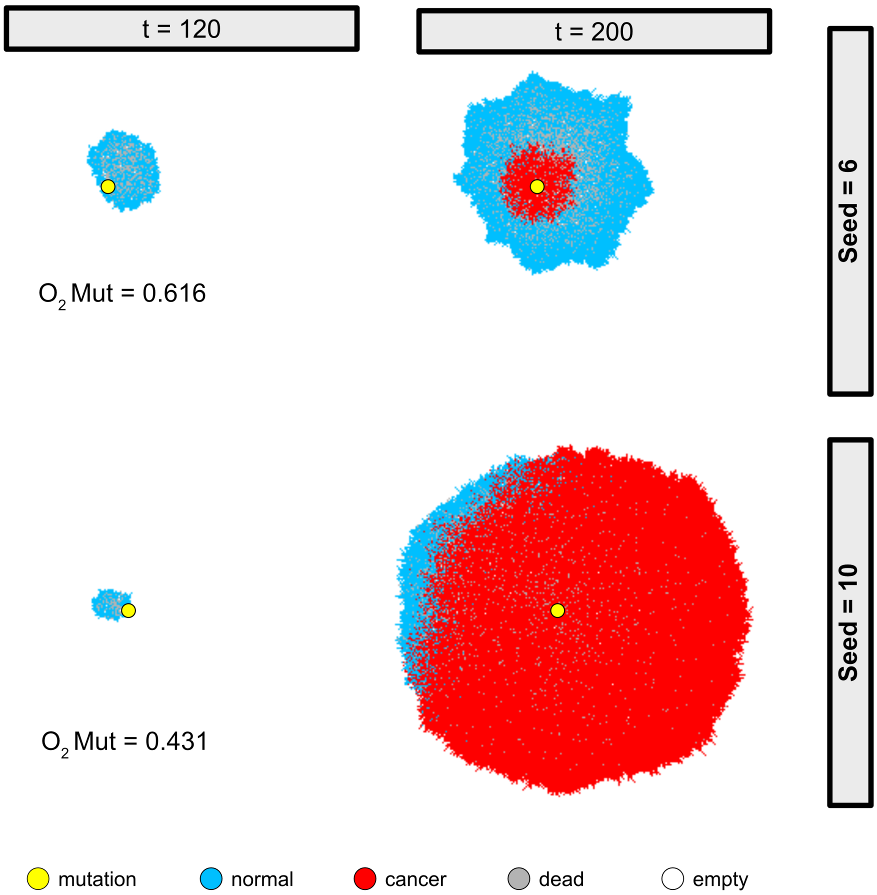

Figure 2: Lattice snapshots for seeds 6 and 10 at the mutation time $t_{\mathrm{mut}} = 120$ (left) and at $t = 200$ (right). The yellow circle marks the mutation site, and O_2MUT denotes the local oxygen concentration at that location.

4 Conclusion and Outlook

The model shows predictive results in figure1. while here we focus on stochacitcty this is a bascis model that can be further optimizwd with introducing more factors such of nutrients avalibilbty, and other microenvironmental factors and type of mutations where the patheway effected by mutations. the oxygen could also be modified

In this model, the oxygen field is treated as a static, diffusion-based field. Oxygen consumption due to cellular respiration is therefore neglected for simplicity. In future work, the model could be extended to include oxygen uptake by cells, allowing for a more realistic description of physiological conditions in which oxygen is dynamically consumed and therefore not purely statically diffused.